The accuracy of liquid-liquid phase transition temperatures determined from semiautomated light scattering measurements

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The synthetic-method determination of liquid-liquid coexistence curves using semiautomated light scattering instrumentation and stirred samples is based on identifying the coexistence curve transition temperatures (T_{cx}) from sudden changes in turbidity associated with droplet formation. Here we use a thorough set of such measurements to evaluate the accuracy of several different analysis methods reported in the literature for assigning T_{cx}. More than 20 samples each of weakly opalescent isobutyric acid+water and strongly opalescent aniline+hexane were tested with our instrumentation. Transmitted light and scattering intensities at 2°, 24°, and 90° were collected simultaneously as a function of temperature for each stirred sample, and the data were compared with visual observations and light scattering theory. We find that assigning T_{cx} to the onset of decreased transmitted light or increased 2° scattering has a potential accuracy of 0.01 K or better for many samples. However, the turbidity due to critical opalescence obscures the identification of T_{cx} from the light scattering data of near-critical stirred samples, and no simple rule of interpretation can be applied regardless of collection geometry. At best, when 90° scattering is collected along with transmitted or 2° data, the accuracy of T_{cx} is limited to 0.05 K for near-critical samples. Visual determination of T_{cx} remains the more accurate approach in this case. © *2010 American Institute of Physics*. [doi:10.1063/1.3469778]

I. INTRODUCTION

Characterizing the solubility behavior of liquid-liquid mixtures has cross-disciplinary impact, from practical problems in chemical engineering to fundamental questions regarding the nature of matter. As an example of the latter case, precise determination of liquid-liquid phase boundaries has played a key role in verifying renormalization group theory predictions for the values of power law exponents,[1–4] thus supporting the concept of critical point universality.[5–7] As an example of the former case, the potential benefits of ionic liquids as solvents for green chemistry has stimulated several recent studies of ionic liquid miscibility.[8–14]

The one-phase/two-phase equilibrium boundaries of liquid-liquid mixtures have been determined using a variety of experimental techniques.[3,15–18] With the synthetic-method, the coexistence curve is constructed by first preparing multiple mixtures spanning a range of compositions, and then measuring the one-phase/two-phase transition temperature of each mixture independently. The formation of droplets of the second phase will cause a stirred binary liquid mixture to become turbid. Consequently, optical techniques are commonly employed in the synthetic-method to observe these phase transitions. The temperature at which turbidity can first be identified, either by eye or with the aid of electronic light detection, is referred to as the cloud point.

Complete determination of the coexistence curve is a time-consuming process: establishing a precise equilibrium transition temperature can take hours for each data point. Efforts to streamline coexistence curve data collection[8,19–30] have been based almost entirely on the synthetic-method for two reasons. First, the optical approach is less invasive. A sample can be prepared in a transparent sealed container, thus limiting impurity accumulation and preventing the system composition from changing over time. Second, sample testing is readily automated. In the prototypical experiment, the optical signal is collected repeatedly while ramping or stepping the sample temperature. Aside from the initial sample preparation, all data can be collected using straightforward computer control of instrumentation for temperature regulation, temperature measurement, and optical signal detection. The majority of semiautomated instruments described in the literature measure transmitted light intensity as a function of temperature; a few instruments monitor off-axis light scattering instead.

Compositional accuracy and precision in these semiautomated experiments are reported to a few parts in ten thousand, which is consistent with analytical balance measurements of gram-scale samples. The accuracy and precision commonly reported for temperature measurements are on the order of 0.01 K. This limit is dependent on well-known experimental perturbations including sample impurities,[3,31] pressure,[32–34] stirring,[35–38] laser heating,[39,40] and gravity.[3,41–43] Less attention has been paid to the accuracy associated with *ex post facto* determination of the phase transition temperature from the recorded optical signal. The implicit expectation is that the transition temperature can be clearly identified by a decrease in transmitted intensity as the sample undergoes phase separation. Several different approaches to analysis of the optical signal have been reported.

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The phase transition temperature has been identified by (1) the onset of signal decline,[24] (2) a set percentage loss of the transmitted intensity,[19–21] (3) complete loss of the transmitted intensity,[44] (4) the intersection of linear fits to the transmitted intensity before and during the decline,[25,45] and (5) analysis using Ornstein–Zernike–Debye (OZD) scattering theory.[28,30,46] The latter method is only applicable to critical mixtures. With instruments which measure light scattered at 45° rather than transmitted light,[8,23,29,47,48] phase transition temperatures have been assigned to the onset of increased signal intensity.

After constructing our own semiautomated liquid-liquid coexistence curve instrumentation and measuring multiangle light scattering signals from two different systems, we have concluded that these assignment methods are flawed when universally applied to a set of samples spanning an entire phase diagram. We propose that errors arise due to misinterpretation of the light scattering signal. Some of the analysis methods listed above are accurate to 0.01 K for part of the phase diagram, but not the entire diagram; others have no theoretical basis and will not be accurate to 0.01 K for any region of the phase diagram. Depending on the analysis method, the magnitude of the assignment error can range from a few hundredths of a kelvin to more than 1 K. Furthermore, the magnitude of the temperature assignment error for a particular set of phase diagram data will not be constant across the phase diagram: the error will vary with composition because the nature of the light scattering signal depends on a sample's proximity to the critical composition.

The intent of this article is to establish best practices for data analysis when liquid-liquid phase transition temperatures are measured from stirred samples using a semiautomated light scattering instrument, and to establish the accuracy that may be expected. We begin with a brief overview of the mechanisms by which phase separation takes place in a binary liquid system, and then discuss the different ways in which a liquid mixture scatters light. Next we describe our instrumentation and methods. Finally, we describe the correlations we have observed between light scattering signal intensities and phase transition temperatures and discuss the impact of these correlations on the accuracy of the various approaches to data interpretation that have been reported.

II. LIGHT SCATTERING FROM LIQUID-LIQUID MIXTURES

The thermodynamic description of liquid-liquid mixtures is well developed (see Refs. 16, 49, and 50, and references therein). Figure 1 shows a generic coexistence curve found in ≈40% of systems exhibiting partial miscibility.[51] This particular coexistence curve has a critical temperature T_C above which the system is completely miscible, and below which the system may spontaneously separate into two phases. The coexistence curve in Fig. 1 encompasses two regions of instability. A single-phase system is completely unstable within the spinodal curve boundary[52,53] and metastable in the regions between the coexistence curve and the spinodal curve.

If a single-phase mixture is prepared near the coexistence curve, then a small temperature quench will take the mixture into either the metastable region (arrow 1 in Fig. 1)

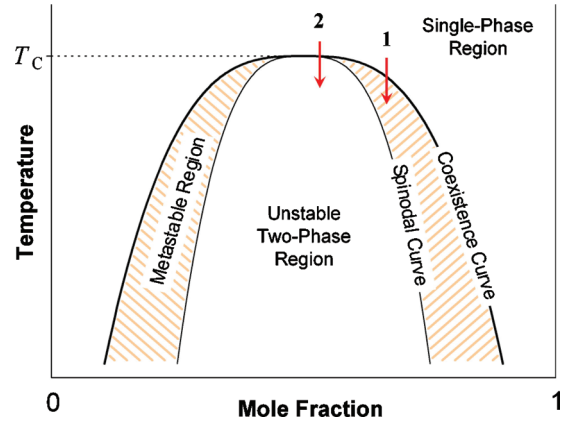

FIG. 1. A representative temperature-composition phase diagram for a partially miscible liquid-liquid mixture with an upper critical point. Arrow 1: Phase separation by droplet nucleation and growth. Arrow 2: Phase separation by spinodal decomposition.

or the unstable region (arrow 2). The phase separation processes are different in these two regions. Phase separation in the metastable region occurs through a process of nucleation and growth,[54,55] where discrete, tiny droplets of the second phase form, grow by solute diffusion, and combine.[56] If a liquid-liquid mixture is free of mechanical agitation and heterogeneous nucleation sites, then significant supercooling can take place.[54,57–61] In order for a cloud point measurement to correspond to a point on the coexistence curve, the experiment must be designed to avoid metastability, and mechanical stirring is usually sufficient to do this.[58,62] However, vigorous stirring can also induce metastability, as will be discussed later.

In contrast to jumps into the metastable region, temperature quenches of near-critical samples, such as arrow 2 in Fig. 1, do not immediately lead to the formation of droplets. Instead, phase separation takes place by a different process known as spinodal decomposition.[52,54,57] In the unstable region of the phase diagram, molecular diffusion occurs *against* the concentration gradient, and domains of higher and lower concentrations develop spontaneously across the entire fluid all at once. For sufficiently small temperature jumps, critical slowing down can cause the time scale for spinodal decomposition in liquid-liquid mixtures to be on the order of minutes.

Both methods of phase separation increase the turbidity τ of a stirred sample, which in turn decreases the intensity I of light transmitted through the sample,

$$I = I_0 \exp(-\tau l). \tag{1}$$

In Eq. (1), I_0 is the incident light intensity and l is the optical path length. The total sample turbidity will include contributions from the homogeneous bulk solution as well as from droplets once they have formed:

$$\tau_{\text{total}} = \tau_{\text{droplet}} + \tau_{\text{bulk}}. \tag{2}$$

The temperature at which a discontinuity in turbidity occurs due to droplet formation, the cloud point temperature, is assumed to coincide with the coexistence curve temperature. Here we summarize theoretical predictions for the temperature and angular dependence of both droplet and bulk scat-

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tering in order to assess experimental design and data analysis methods later on. Further details regarding light scattering theory can be found in the supplementary material.[63]

Elastic light scattering theory for spherical droplets is well established.[64–69] If no multiple scattering takes place, then the turbidity for scattering by droplets is

$$\tau_{\text{droplet}} = NC, \tag{3}$$

where N is the droplet number density and C is the droplet scattering cross section. The value of C increases by ten orders of magnitude when the droplet radius undergoes a two order of magnitude increase from 10 nm to 1 μm.[63] This is the basic premise behind the cloud point technique: the light scattering signal changes abruptly at the coexistence curve temperature T_{cx} when τ_{droplet} suddenly becomes nonzero. The rate of change of τ_{droplet} with temperature will be smaller for isopleths where the slope of the coexistence curve $|\mathrm{d}T_{\text{cx}}/\mathrm{d}x|$ is steep because the droplet number density in Eq. (3) will change more slowly with temperature due to lever rule considerations (Appendix).

The second contributor to Eq. (2), τ_{bulk}, arises from spontaneous composition and density fluctuations in the bulk liquid solution. Once droplets form, τ_{droplet} is usually much greater than the baseline turbidity from the bulk. However, concentration fluctuations become very large on the spinodal curve (Fig. 1), making τ_{bulk} appreciable in the immediate vicinity. This behavior is known as critical opalescence and is described by OZD theory.[70–75] Unlike the droplet light scattering signal, OZD scattering does not have a sudden temperature onset, and it occurs both above and below T_{cx}. Bulk liquid scattering intensity from a sample of critical composition builds continuously as $T \rightarrow T_C$, and then declines as the sample passes into the two-phase region, where the compositions of the separated phases move away from the spinodal curve (Fig. 1). An off-critical single-phase sample that is still in the flattened region of the coexistence curve will also be opalescent, albeit with less intensity. The turbidity for such a sample will peak at T_{cx} unless the sample supercools, in which case the opalescence will continue to grow as the single-phase system moves into the metastable region and on toward the spinodal curve. The contribution of OZD scattering from the bulk solution has often been overlooked in semiautomated coexistence curve measurements, and we will show that OZD scattering has a substantial impact on correct assignment of the cloud point temperature. For both droplet and bulk contributions, the relative strength of scattering depends on the refractive indices of the pure liquids. The greater the disparity in these indices, the stronger the droplet and OZD scattering.[63]

The angular distribution of droplet and bulk light scattering is also of importance to the experiment. With droplet scattering, the scattered intensity distribution shifts to the forward direction as the droplet size increases. For example, more than half of the light scattered from a 1 μm droplet is predicted to be within 8° of the incident axis, and this angle will narrow further as the droplet grows.[63] Therefore, instrumentation based on scattered light (rather than transmitted light) should be designed to collect scattered light close to the incident axis in order to achieve the greatest experimental sensitivity for droplet formation. With OZD scattering, spatially correlated composition fluctuations in the liquid will introduce a degree of coherence into the scattered light signal, leading to slight anisotropy in the angular distribution. However, the asymmetry is orders of magnitude less pronounced[76–78] compared to the angular distribution from all but the smallest of droplets. For the purposes of this experiment, OZD scattering can be treated as isotropic.

TABLE I. Refractive indices (n_{D}) and densities (ρ) of isobutyric acid, water, aniline, and hexane (Ref. 92).

Compound	n_{D}	ρ (g/ml)	T (°C)
Isobutyric acid	1.393	0.968	20
Water	1.333	0.998	20
Aniline	1.586	1.022	20
Hexane	1.373	0.661	25

In addition to droplet and bulk scattering, there is a unique transient light scattering phenomenon that occurs when an unstirred critical solution undergoes spinodal decomposition (arrow 2 in Fig. 1). The resulting diffusion against the concentration gradient that occurs everywhere in the solution creates alternating regions of higher and lower concentration that grow in size with time.[52,57] Because of this spontaneously generated periodicity, light incident on a solution in the midst of spinodal decomposition scatters in a characteristic ring pattern similar to x-ray or electron diffraction powder patterns. As the domains grow in size over time, the ring intensity increases and the ring diameter collapses.[39,79–83] With a small enough jump across a binary liquid coexistence curve, the growth process can last for minutes due to critical slowing down. In our experience, the ring is readily observed with 0.01 K temperature control, and therefore provides unequivocal confirmation of phase transition temperatures near the critical point.

III. EXPERIMENTAL DETAILS

Our objectives were to (i) collect light scattering signals at different angles from a range of liquid-liquid samples taken slowly through their phase transitions, (ii) to compare these signals with direct visual observation of the phase transitions, and (iii) to then evaluate the different methods of data analysis listed earlier in the introduction. To do this, light signals from four different scattering angles were collected simultaneously. Three of the detectors were positioned at angles commonly used by other researchers: transmitted, small angle, and off axis. A fourth detector was positioned at 90° and optimized for the detection of critical opalescence.

Two binary systems with markedly different physical characteristics were chosen for this investigation. Isobutyric acid and water have well-matched refractive indices and densities (Table I). The system is weakly opalescent and has an upper critical point near room temperature. In contrast, aniline and hexane have disparate densities and refractive indices, at least compared to the range of values commonly

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found in liquids subjected to coexistence curve analysis. The system is strongly opalescent and has an upper critical point near 340 K.

A. Materials and sample preparation

Puriss-p.a.-grade isobutyric acid (99.9%) was purchased from Fluka. Lot analysis by Fluka reported the presence of 0.12% water by mass. Puriss-p.a.-grade water suitable for inorganic trace analysis was purchased from Fluka. ACS-reagent-grade aniline (99.9%) was purchased from Aldrich and stored in a freezer. The aniline was colorless. Puriss-p.a.-grade hexane standard for GC (99.7%) was purchased from Fluka. Lot analysis by Fluka reported that the impurities were 2-methylpentane (0.07%) and methylcyclopentane (0.23%) with water content $<0.01\%$. All chemicals were used without further purification beyond drying the aniline and hexane over molecular sieves.

Samples were prepared in custom flat-bottomed, dome-topped borosilicate ampules (Chemglass) with nominal internal volumes of 2.5 ml. Each ampule contained a Teflon-coated magnetic spin bar. Pure components were transferred by syringe, dissolved gases were removed from the mixture with freeze-pump-thaw cycles, and the ampules were flame-sealed. Sample compositions were determined gravimetrically and corrected for buoyancy and evaporation to the head space. Further details regarding sample preparation are in the supplementary information.[63]

Twenty one aniline+hexane samples were prepared. Aniline degrades under exposure to oxygen and light,[84,85] but the equilibrium phase transition temperatures of these samples remained consistent to within the temperature uncertainty of our instrument after over a year of cold storage. Forty-one isobutyric acid+water samples were created. Isobutyric acid+water mixtures are known to exhibit long-term drifts in phase transition temperatures, even in flame-sealed ampules.[38,86,87] Proposed reasons for these drifts include the leaching of impurities from the container walls and the growth of microorganisms. We observed that the values of T_{cx} for many of our samples decreased on the order of 0.1 K within the first 2 months. In most cases the values of T_{cx} then remained stable over the next 2 years, although samples on the water-rich edge of the coexistence curve showed more pronounced and prolonged declines in T_{cx}. We also noted the formation of trace amounts of a solid-like third phase following phase separation of near-critical isobutyric acid+water samples, which is consistent with observations reported by Jacob *et al.*[88] The phase forms at the meniscus between the two layers, but its presence did not interfere with the light scattering measurements.

B. Temperature control and measurement

Light scattering measurements on samples were carried out inside a custom unsilvered glass Dewar flask manufactured by Chemglass. Design details are in the supplementary material.[63] The temperature of the recirculating fluid (water) was controlled with a refrigerated bath circulator (Neslab, RTE-111), programmable to 0.01 K precision. The ampule under study was held inside a stem piece that was easily removed from the Dewar without disturbing the optics, and we could exchange ampules in under 2 min with highly reproducible placement in the optical path. The ampule spin bar coupled over a 3 cm distance to a spin bar in the bottom of the Dewar. Stirring rates were measured with a stroboscope. The maximum stable stirring rate was 10 Hz, and this rate was used during most light scattering measurements.

The temperature inside the Dewar was measured with a thermistor (Omega, ON-410-PP) and handheld meter (Omega, HH42). The sensor tip of the thermistor was positioned at the vertical midpoint of the ampule body, and only a few millimeters away from the outer wall of the ampule. Calibration of the thermistor/meter pair was NIST-traceable. The overall accuracy of our temperature readings was estimated to be ± 0.02 K.[63]

C. Optics

Samples were illuminated with 3 mW of power from a linearly polarized (150:1) 635 nm laser diode (Thorlabs, CPS196). Tests showed no discernible sample heating after exposure to the laser light. The laser's polarization vector was normal to the optical breadboard, and light scattering intensities were collected in the plane of the breadboard. Scattered light was detected at four angles (transmitted, 2°, 24°, 90°) using amplified photodetectors (Thorlabs, PDA520). The photodetectors responded linearly over the full range of power to which they might be exposed. Spinodal decomposition patterns were observed on a white card placed in front of the optics leading to the transmitted and 2° detectors. A schematic of the instrumental optics and additional details are found in the supplementary material.[63]

The 90° detector optics were designed specifically for the observation of critical opalescence. At this angle, droplet scattering intensities are greatly diminished. The optical path in front of the 90° detector contained a rectangular aperture (Ealing, 61–1137) which was adjusted to a narrow horizontal slit transmitting just the image of the laser beam in the ampule. In this way, light reaching the detector from any other regions of the ampule was blocked; such light would be due to multiple scattering from either droplets, opalescence, or some extraneous source. The rectangular aperture did not eliminate multiple scattering sources in the horizontal plane containing the laser beam.

Initial alignment of the optics was done using a critically opalescent sample in order to provide a strong image of the laser path through the ampule. This process was necessary only once in a great while: the removable stem design of the Dewar assembly allowed us to exchange samples quickly without significantly disturbing optical alignment. The long-term stability of the optical arrangement was very valuable, as it allowed us to make comparative studies of scattering intensities across a full set of samples.

D. Data collection

A practical experimental time scale for sample equilibration had to be established in order for the cloud point temperature to match T_{cx}. We used reported observations of unstirred mixtures with unusually slow diffusion conditions[89,90]

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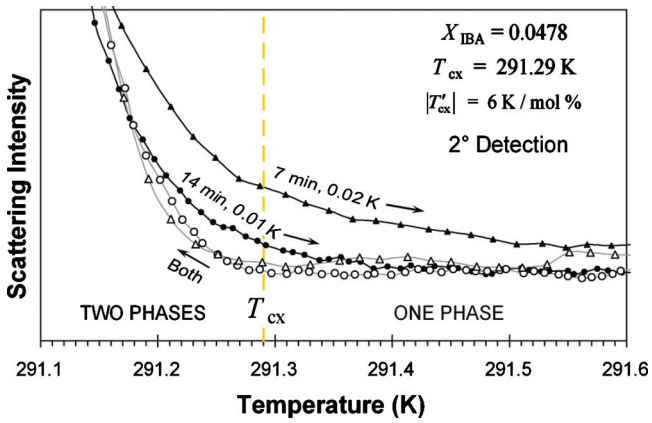

FIG. 2. Temperature dependence of 2° light scattering from stirred isobutyric acid+water (x_{IBA}=0.0478). (○) 14 min equilibration time, ΔT_{Step}=−0.01 K; (●) 14 min, ΔT_{Step}=+0.01 K; (△) 7 min, ΔT_{Step} =−0.02 K; (▲) 7 min, ΔT_{Step}=+0.02 K.

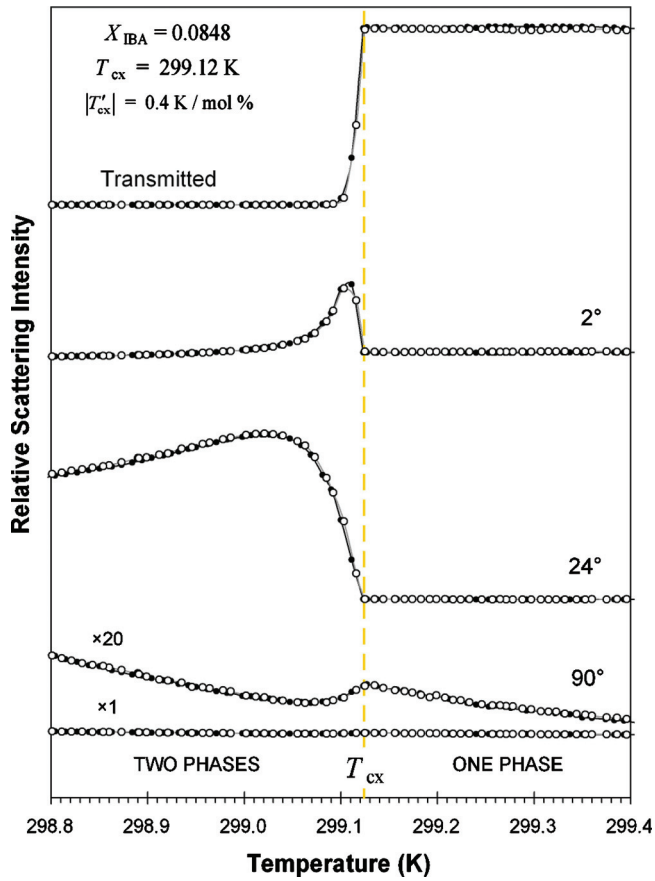

FIG. 3. Temperature dependence of light scattering (transmitted, 2°, 24°, and 90°) from stirred isobutyric acid+water (x_{IBA}=0.0848). (○) ΔT_{Step} =−0.01 K; (●) ΔT_{Step}=+0.01 K.

to set an initial upper limit for the time scale of droplet nucleation and growth. The temperature was changed stepwise in 0.01 K increments, and the stirred sample equilibrated for several minutes at each fixed temperature. In most other semiautomated instrumentation, the temperature is ramped at rates on the order of 10 mK/min (our equivalent "rate" was ≈1 mK/min). With either approach, the presence of hysteresis between runs of decreasing and increasing temperature is a strong indicator that a slower rate or longer equilibration time is necessary. We only observed reproducible hysteresis in samples with large $|dT_{cx}/dx|$, and equilibration times for those samples were increased as necessary.

Data collection for a particular sample began with a quick, crude identification of the phase transition temperature by eye. The computer was then programmed to change the temperature of the sample by 0.01 K steps over a 0.6 K range that encompassed the observed phase transition. The sample was stirred continuously at 10 Hz. To collect one light scattering data point, the bath temperature was first changed by 0.01 K and the sample was given 7 min to equilibrate. Background detector readings were then averaged over a 1 min period, the shutter opened to let the laser into the sample, and the four detector readings were collected and averaged again for another minute. The program then started the next step by incrementing the temperature. A typical run involved two passes: from the single-phase region into the two-phase region, and then from the two-phase region back into the single-phase region. Light scattering signals from the two passes superimposed upon one another to within 0.01 K. There might be slight variations in relative intensity, but qualitative variations of intensity with temperature lined up perfectly in most cases (Figs. 3–5). If the phase transition temperatures identified by the two passes disagreed by more than 0.01 K, then the sample was rerun with a longer equilibration time. The time required for a complete run of a single sample was on the order of 12–18 h. Runs generally took place overnight with the instrument unattended.

At the end of a run, identification of the phase transition temperature was confirmed by eye. The sample was allowed to equilibrate in the single-phase region a few hundredths of a kelvin above the expected transition temperature. After an 0.01 K step, the sample was stirred for several minutes, the stirring was turned off, and then the sample was observed for several more minutes with and without the laser. This process was repeated until either (i) droplets appeared, (ii) distinct macroscopic regions differing in refractive index were evident when stirring was resumed after several minutes of settling, or (iii) the spinodal decomposition diffraction ring pattern appeared (for samples very near the critical composition). We also monitored the electronic detector signals during this secondary evaluation process, and the intensities as a function of temperature agreed with the averages collected during the overnight run.

All light scattering signals exhibited long-term reproducibility. Absolute scattering intensities varied somewhat due to modifications to the optics, and the absolute location of T_{cx} changed over the course of 2 years in many of the isobutyric acid+water samples. However, for a given isobutyric acid +water sample in which T_{cx} did drift, the light scattering signal as a function of temperature drifted along with T_{cx}. In other words, the light scattering graphs plotted *relative* to T_{cx} in Fig. 6 did not change over time, even though the actual value of T_{cx} itself might have changed.

IV. EVALUATION OF LIGHT SCATTERING SIGNALS

Here we show how interpretation of the light scattering signal depends on the slope of the coexistence curve at a given sample composition, and correlations are made be-

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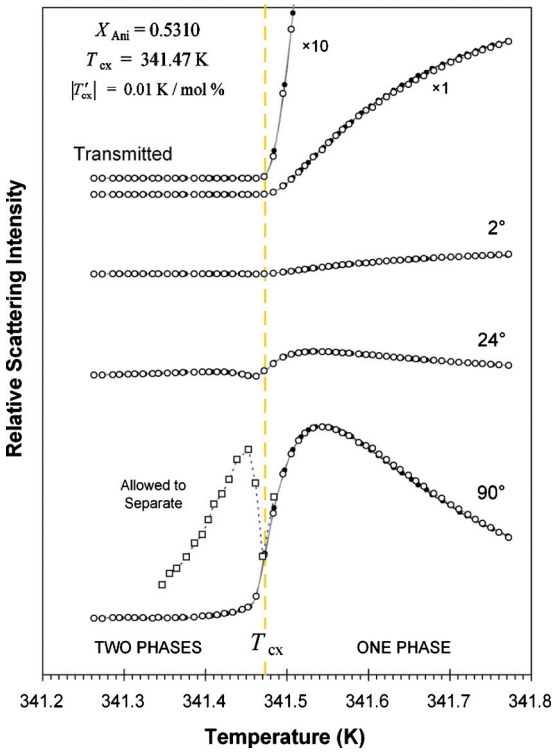

FIG. 4. Temperature dependence of light scattering (transmitted, 2°, 24°, and 90°) from stirred aniline+hexane (x_{Ani}=0.5310). (○) ΔT_{Step}=−0.01 K; (●) ΔT_{Step}=+0.01 K; (□) ΔT_{Step}=−0.01 K and sample settled for 10–20 min with no stirring before measuring scattered light.

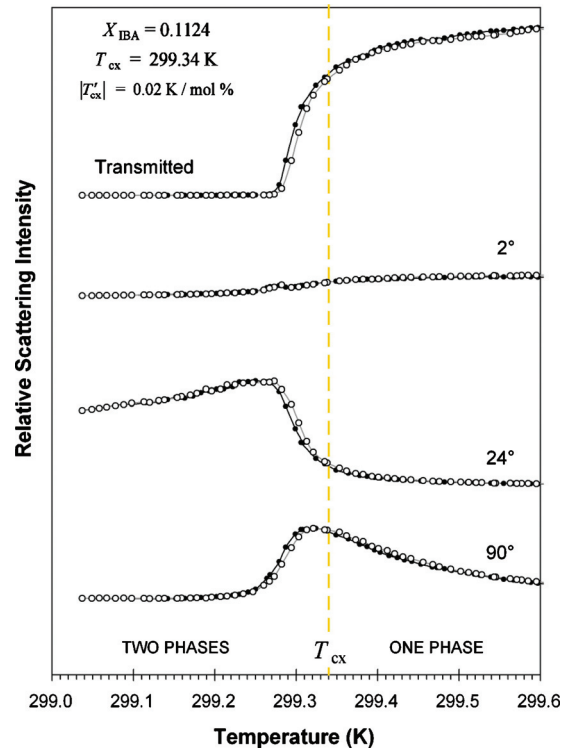

FIG. 5. Temperature dependence of light scattering (transmitted, 2°, 24°, and 90°) from a stirred isobutyric acid+water (x_{IBA}=0.1124). (○) ΔT_{Step} =−0.01 K; (●) ΔT_{Step}=+0.01 K.

tween the T_{cx} values determined by eye with the temperature-dependent light scattering signals for each of the four collection angles (transmitted, 2°, 24°, and 90°). To shorten the notation, we will abbreviate the slope $|dT_{cx}/dx|$ as $|T'_{cx}|$. The general light scattering behavior is broken into three categories using $|T'_{cx}|$ as a semiquantitative measure. The numerical boundaries dividing these categories are fuzzy and based only on our own empirical observations. Values of $|T'_{cx}|$ were determined using polynomial fits of the coexistence curves and are reported in units of K/mol % for convenience of interpretation and representation.

Selected light scattering data are shown in detail in Figs. 2–5 and will be discussed further below. To aid the discussion, particularly with respect to correlations between light scattering signals and $|T'_{cx}|$, multiple additional data sets are presented in condensed format in Fig. 6 (11 isobutyric acid +water mixtures) and Fig. 7 (11 aniline+hexane mixtures). The rows in these figures correspond to different sample compositions, progressing downward in order of increasing mole fraction of isobutyric acid or aniline. The leftmost column of each figure highlights the location of a given mixture on the temperature/composition phase diagram. The second column reports the value of $|T'_{cx}|$ at that composition. The remaining four columns in Figs. 6 and 7 present an array of minigraphs that show the dependence of light scattering data on temperature, sample composition and collection angle. The temperature axis of each minigraph is plotted relative to T_{cx} for that sample and spans about 0.4 K around T_{cx}. All minigraphs within a given column are then aligned by T_{cx} so that scattering signals in the one-phase and two-phase regions may be compared as a function of composition. The vertical axes for these minigraphs are relative scattering intensities; no attempt was made to measure absolute scattering intensities. The rigid design of our light scattering instrumentation ensured reproducible signal strengths even though data collection took weeks. Therefore, all graphs within a given column are on the same relative intensity scale and may be compared directly to one another.

A. Samples with $|T'_{cx}|$>1 K/mol %

Samples with large $|T'_{cx}|$ are well-removed from the critical point, and so $\tau_{droplet} \gg \tau_{bulk}$ in a two-phase mixture [Eq. (2)]. Measurements of transmitted intensity versus temperature show a discontinuity in slope at T_{cx} due to the onset of droplet scattering. The slope of declination in the two-phase region will depend on the rate at which the extensive amount of nascent phase forms with respect to temperature. This rate is inversely proportional to $|T'_{cx}|$ [Eq. (A3)]: the steeper the coexistence curve, the slower the growth of the light scattering signal. This effect can be seen by examining the transmitted intensity graphs in Figs. 6 and 7. The transmitted intensity declines slowly on either side of the aniline +hexane coexistence curve (x_{Ani}=0.1634,0.8022) and on the steep water-rich side of the isobutyric acid+water coexistence curve (x_{IBA}=0.0478). In both cases, the rate of declination accelerates rapidly as $|T'_{cx}|$ decreases.

Scattering data at 2° are the inverse of transmitted data: the location of T_{cx} correlates to an increase in off-axis scattering intensity caused by droplet formation. In our experience, the 2° scattering geometry is more sensitive than the transmitted geometry to droplet formation in samples with

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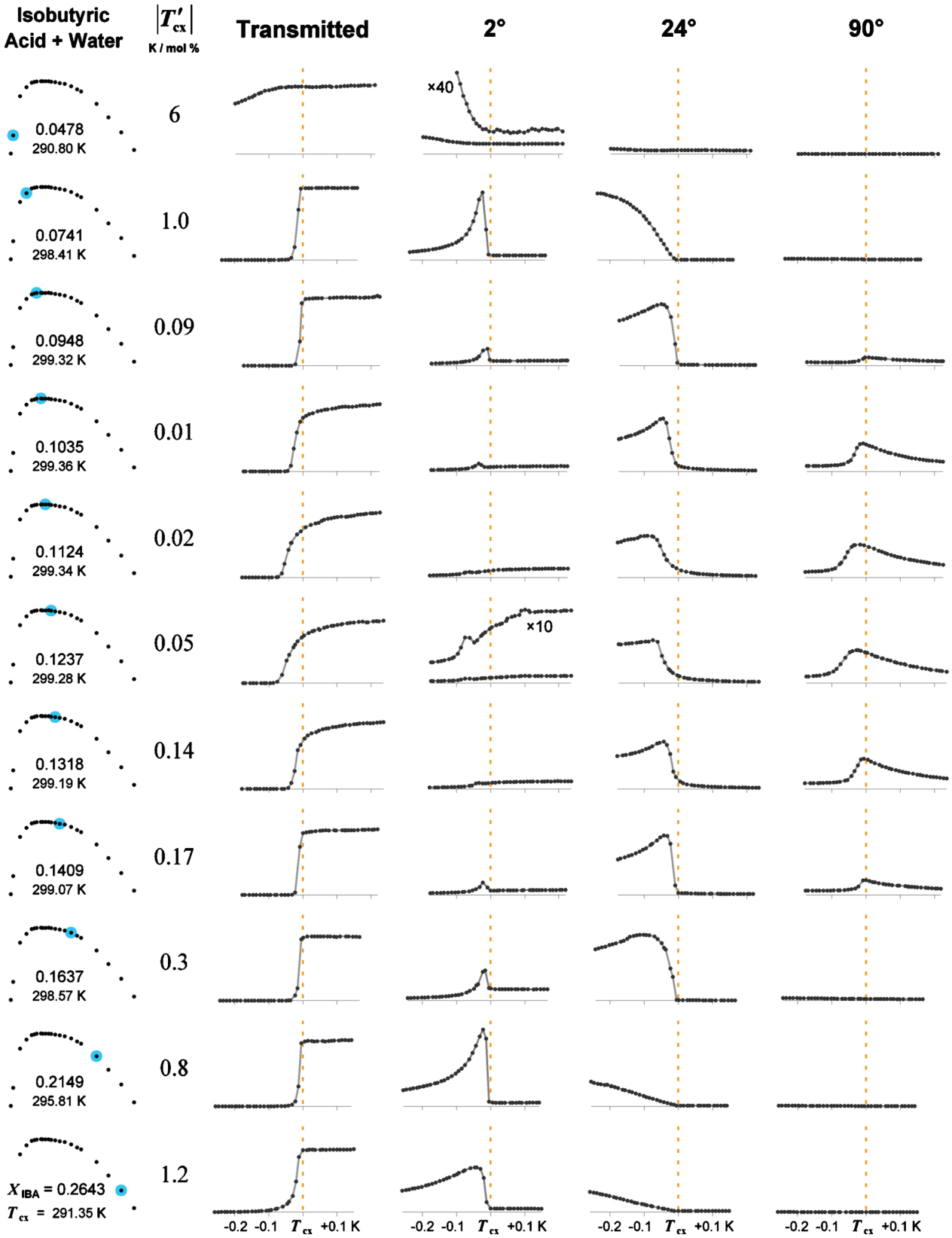

FIG. 6. Temperature and composition dependence of light scattering (transmitted, 2°, 24°, and 90°) for 11 stirred isobutyric acid+water mixtures. Rows: increasing x_{IBA} composition; column 1: sample location on phase diagram; column 2: magnitude of coexistence curve slope (K/mol %); columns 3–6: scattering intensity vs temperature and collection angle. Temperature axes are plotted relative to the T_{cx} value for each sample.

large $|T'_{cx}|$. Weak scattering intensity grows on a near-zero baseline in the 2° data, whereas with the transmitted intensity data, weak scattering causes a small decrease of a large baseline signal. For most samples with $|T'_{cx}| > 1$ K/mol %, the 2° scattering data also revealed the phase transition more clearly than data collected at 24°; no indications of a phase transition were evident in the 90° signals collected from these samples.

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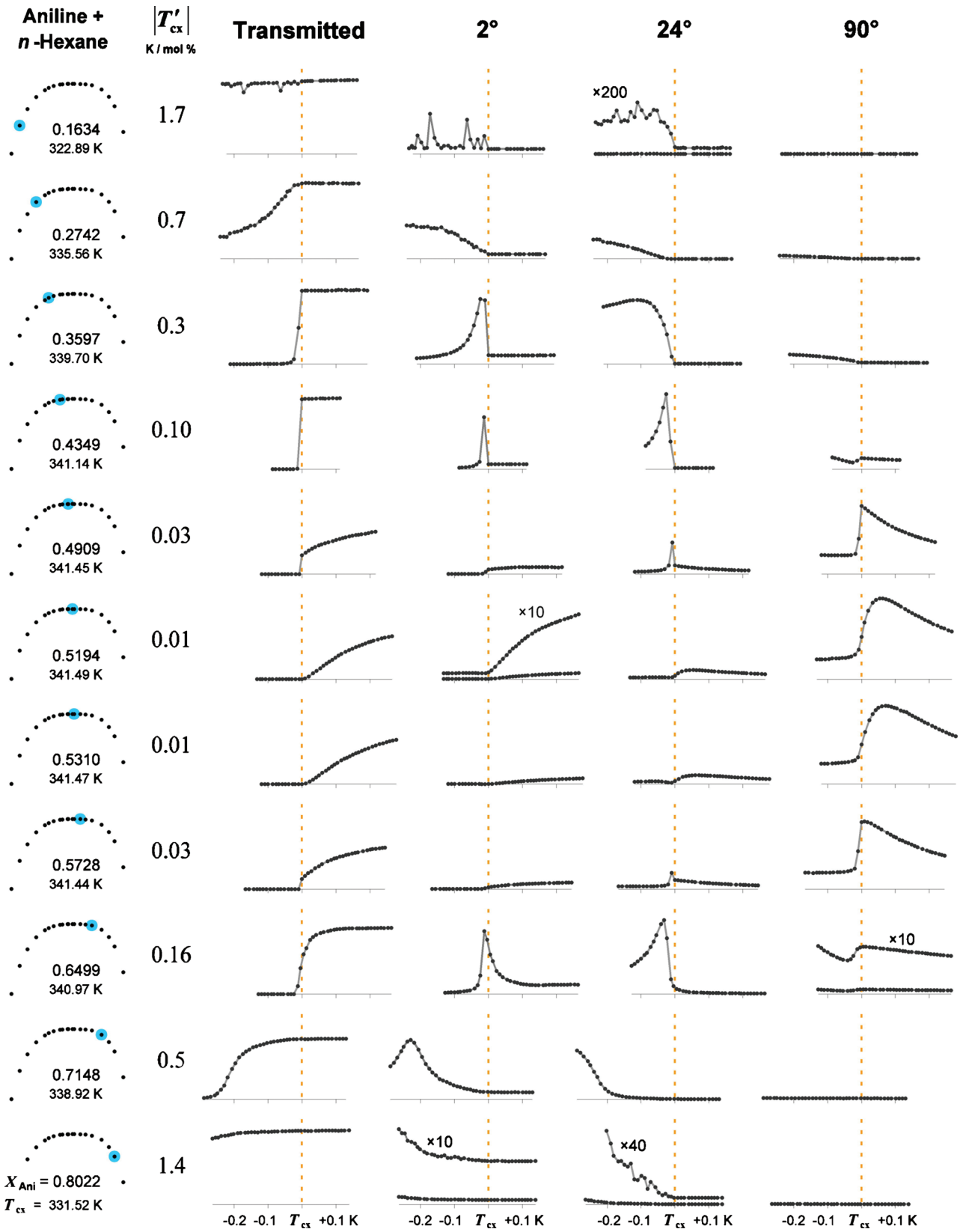

FIG. 7. Temperature and composition dependence of light scattering (transmitted, 2°, 24°, and 90°) for 11 stirred aniline+hexane mixtures. Rows: increasing x_{Ani} composition; column 1: sample location on phase diagram; column 2: magnitude of the coexistence curve slope (K/mol %); columns 3–6: scattering intensity vs temperature and collection angle. Temperature axes are plotted relative to the T_{cx} value for each sample.

We also observed that hysteresis can be a significant problem for samples with large $|T'_{\mathrm{cx}}|$. As seen in Figs. 3–5, scattering signals from runs of increasing and decreasing temperature superimpose almost perfectly for mixtures with $|T'_{\mathrm{cx}}|<1$ K/mol %. These data were collected with 7 min of equilibration time after each 0.01 K step. In contrast, Fig. 2 shows a magnification of 2° scattering signals from an isobutyric acid+water sample with $|T'_{\mathrm{cx}}|=6$ K/mol %

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($x_{IBA}=0.0478$). The apparent phase transition observed in a run of increasing T (2-phase→1-phase) occurs at a higher temperature than in a run of decreasing T (1-phase→2-phase). Our initial concern was that the single-phase mixture was prone to supercooling. To test this, we ran multiple runs with varied equilibration times and step sizes. Significantly less equilibration time, or larger step sizes, caused both apparent phase transition temperatures to move even further apart. However, as seen in Fig. 2, simultaneously doubling the equilibration time and halving the step size did not alter the scattering signal for runs of decreasing temperature, but did reduce the hysteresis relative to the corresponding run of increasing temperature. We conclude from this that runs of decreasing temperature were not supercooled; instead, this particular two-phase sample required more time to mix thoroughly in order to fully equilibrate.

Other samples show the opposite behavior. On the isobutyric-acid-rich side of the phase diagram, insufficient stirring time caused the 2-phase→1-phase transition to occur at a lower temperature than the 1-phase→2-phase transition. Again, increasing the equilibration time moved $T_{cx,2\to1}$ toward $T_{cx,1\to2}$, but $T_{cx,1\to2}$ did not change. These observations of hysteresis demonstrate the importance of evaluating light scattering data using runs of both increasing and decreasing temperatures. In addition, our observations show that when hysteresis is present in the data, assigning T_{cx} as the average of the results from increasing and decreasing temperature trials is a questionable practice.

B. Samples with $|T'_{cx}|\cong 0.4$ K/mol %

Light scattering data from samples with coexistence curve slopes on the order of 0.4 K/mol % were the most straightforward to interpret. The nascent phase forms rapidly with changing temperature [Eq. (A3)], and the droplet scattering that results is much stronger than weak OZD scattering. As seen in Fig. 3 for $x_{IBA}=0.0848$, there is an immediate increase in the amount of light scattered at 2° once the temperature falls below T_{cx}, corresponding to the nucleation and growth of the second phase. At the same time, the transmitted intensity abruptly decreases because the droplets scatter light off incidence.

In Fig. 3, when the temperature is only 0.02 K below T_{cx}, the turbidity due to the number and cross section of the droplets is so great that the transmitted intensity is negligible. This high level of turbidity indicates that multiple scattering is taking place, and as a result the 2° scattering intensity decreases with temperature almost as quickly as it grows in. As the temperature falls further below T_{cx}, multiple scattering spreads the incident light to wider and wider angles: the scattering intensity at 24° grows in more slowly than the 2° scattering intensity and reaches a maximum at a lower temperature. Even the 90° signal shows a slow growth in intensity with decreasing temperature. Other samples exhibiting this type of behavior include $x_{IBA}=0.1637$ and 0.2149 in Fig. 6 and $x_{Ani}=0.3597$ in Fig. 7.

Droplet scattering explains most of the characteristics present in the data in Fig. 3, but does not account for the small peak at T_{cx} seen in the 20× magnification of the 90° scattering signal. We attribute this peak to OZD scattering from the bulk solution. Although the $x_{IBA}=0.0848$ sample is removed from the critical point in terms of both composition and temperature, weak critical opalescence still exists in the sample. This weak opalescence peak is not found in magnifications of either the 2° or the 24° signals, but the optics at the 90° collection angle have been optimized for the detection of bulk liquid OZD scattering. As expected, the opalescence reaches its maximum at T_{cx}.

C. Opalescent samples ($|T'_{cx}|<0.2$ K/mol %)

The coexistence curve slope approaches zero as sample composition approaches the critical composition. A sample in this region of the coexistence curve will exhibit OZD light scattering at temperatures near T_{cx}, both as a single-phase and as two phases. To explore the effects of critical opalescence on T_{cx} assignment methods, light scattering data were collected from a strongly opalescent system (aniline+hexane) and a weakly opalescent system (isobutyric acid+water). Aniline and hexane have disparate refractive indices (Table I), so their droplet scattering cross sections and OZD scattering intensities are larger than many other liquid pairs.[63] The strength of OZD scattering from aniline+hexane is seen in the 90° scattering signal from a near-critical $x_{Ani}=0.5310$ solution (Fig. 4). The 90° signal begins building up in the single-phase region many tenths of a degree prior to the T_{cx} identified by eye. In fact, the opalescence is so strong that multiple scattering caused by the high turbidity eventually attenuates the amount of light collected at 90°, and T_{cx} occurs at a temperature several hundredths of a degree lower than the measured peak maximum. There is no signature within the 90° signal to indicate exactly where T_{cx} occurs.

To confirm our visual observations, we carried out a follow-up experiment where the mixture was stirred for 10 min at constant temperature and then allowed to separate for 10–20 min. The laser passed through the upper, hexane-rich layer of the separated sample. The process was repeated over a range of temperatures starting just above the visual determination of T_{cx}. These data are superimposed on the 90° stirred scattering signal in Fig. 4 and show how the hexane-rich layer still exhibits critical opalescence even after phase separation takes place. The point of maximum opalescence and maximum turbidity is T_{cx}, which for this sample corresponds to the *minimum* of the unstirred 90° signal because of multiple scattering. The location of T_{cx} is also clearly the point of departure between the stirred signal and the unstirred signal.

While T_{cx} could be identified by the onset of the decline in the transmitted intensity signal when $|T'_{cx}|$ was on the order of 0.4 K/mol % or larger, strong OZD scattering in near-critical aniline+hexane causes the transmitted light to decline well before the phase transition takes place (Fig. 4). For the $x_{Ani}=0.5310$ sample, T_{cx} matches the point where the transmitted intensity becomes negligible. We believe this is coincidental: the accuracy of this correlation depends on the optical path length of the sample [Eq. (1)]. With a thicker sample, the transmitted intensity would become negligible at

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a temperature above T_{cx}. With a thinner sample, the transmitted intensity would show a gradual decline due to opalescence, followed by an abrupt drop at T_{cx} when the turbidity suddenly increases due to droplet scattering. This latter behavior is seen in light scattering data of Fig. 7 from less opalescent near-critical samples (x_{Ani}=0.4909 and 0.5728).

The behavior of the transmitted light signal in the x_{Ani} =0.5310 sample is echoed in the 2° and 24° scattering data (Fig. 4). The 2° scattering signal actually declines as T_{cx} is approached from the single-phase region because the strong critical opalescence causes multiple scattering and deflects light away from the 2° detector.

In contrast to aniline and hexane, the refractive indices of isobutyric acid and water are closely matched (Table I), and the system is weakly opalescent. As a result, the nature of the light scattering signal from a stirred near-critical isobutyric acid+water sample (Fig. 5) is different than that from near-critical aniline+hexane (Fig. 4) in several ways. First, the transmitted light intensity through x_{IBA}=0.1124 undergoes less decay as the temperature approaches T_{cx} and critical opalescence builds. Second, visual observation of spinodal decomposition puts T_{cx} several hundredths of a kelvin higher than the eventual rapid decline of the transmitted intensity. Third, the effects of droplet scattering manifest more gradually below T_{cx} in near-critical isobutyric acid +water samples because the matching refractive indices and similar compositions of the two phases initially make for weak scattering. As a result of these factors, visual observations of T_{cx} for near-critical samples of isobutyric acid +water mixtures do not correlate to any distinctive change in slope in the transmitted intensity curve (Fig. 5 and x_{IBA} =0.1035–0.1318 in Fig. 6).

The location of T_{cx} is also slightly higher than the maximum in the 90° signal collected, which was initially surprising. However, this assignment was conclusively confirmed by multiple experiments. First, all near-critical isobutyric acid+water samples we made exhibited this behavior. Second, the characteristic spinodal decomposition ring pattern was observed when an equilibrated, *unstirred* sample at our assigned T_{cx} underwent an 0.01 K quench. Third, when a *stirred* sample was allowed to equilibrate 0.01 K below T_{cx}, stopping the stirring caused the sample to undergo spinodal decomposition.

The assignment is also consistent with previously reported observations of liquid-liquid phase transition suppression caused by stirring.[35,37,38] In off-critical liquid-liquid mixtures, the shear force of vigorous mechanical stirring will break up larger droplets during nucleation and growth, and if the stirring is sufficiently rapid then the droplet size distribution can be brought below the threshold size necessary for spontaneous growth. The formation of a second phase will be suppressed. In a similar way, shear forces can inhibit domain growth in near-critical samples during spinodal decomposition. At stirring frequencies of 30 Hz, some phase transitions are suppressed up to 0.05 K away from the coexistence curve boundary.

To check the effects of stirring suppression, we tested two of our isobutyric acid+water samples, one critical and one off-critical, and observed that the phase transition with 10 Hz stirring was 0.01 K lower than when the samples were stirred at 3.3 Hz. These results are consistent with Easwar's experiments on isobutyric acid+water.[38] Critical and off-critical aniline+hexane samples were also tested at 3.3 and 10. Hz, and there was no evidence of phase transition temperature suppression in those samples. We therefore believe that the slight increase in the 90° scattering signal from stirred samples at temperatures just below T_{cx} (Figs. 3 and 6) occurs because stirring inhibits phase separation for 0.01 or 0.02 K.

The expected increase and decrease in critical opalescence about the critical composition are seen by scanning down the columns of 90° scattering data in Figs. 6 and 7. We had anticipated that the assignment difficulties associated with the transmitted and 2° scattering signals would be resolved by collecting 90° scattering data, as T_{cx} should correspond to the maximum of the critical opalescence signal. While this correlation is seen in the data for some samples, the general rule does not hold. With strongly opalescent near-critical aniline+hexane mixtures, self-attenuation puts T_{cx} at a temperature lower than the measured peak maximum. In contrast, with near-critical isobutyric acid+water mixtures, the value of T_{cx} determined by visual observation occurs 0.01–0.02 K higher than the peak maximum because of stirring suppression.

V. ASSESSMENT OF T_{cx} ANALYSIS METHODS

Combining our observations from Figs. 2–7, we reach the key conclusion of this paper for stirred liquid-liquid mixtures: *regardless of the optical collection geometry, there is no single, simple rule for assigning T_{cx} from light scattering data that is accurate to 0.01 K for all compositions*. To support this statement, we will evaluate each of the published T_{cx} assignment methods listed earlier in the introduction. First, the light scattering data presented in Figs. 6 and 7 show that assigning T_{cx} to the start of the transmitted intensity decline is correct for samples with $|T'_{cx}| > \approx 0.4$ K/mol %, where turbidity increases with varying degrees of abruptness due to droplet formation. However, this analysis method is not correct for most samples with $|T'_{cx}| < 0.2$ K/mol % because the onset of OZD scattering causes the transmitted intensity to decrease well before phase separation actually takes place. Values of T_{cx} assigned in this way will err on the single-phase side of T_{cx}; judging from the curvature in the transmitted intensity curves of Figs. 6 and 7, this error would be on the order of tenths of a kelvin or more.

Second, our data show that assignment of T_{cx} to some percentage of declination or to negligible transmission has no general justification. Assignment of T_{cx} by these methods matches visual observations with 0.01 K accuracy only through fortuitous combinations of optical path length, physical properties of the components, and system composition.

Third, for samples with $|T'_{cx}| > \approx 0.4$ K/mol %, identifying T_{cx} from the intersection of linear fits to the single-phase transmitted intensity and two-phase transmitted intensity is equivalent to identifying T_{cx} by the start of the decline in transmitted intensity. However, there is no theoretical jus-

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TABLE II. Coexistence curve temperatures (T_{cx}) as a function of mole fraction aniline (x_{Ani}) for aniline+hexane liquid-liquid mixtures. Values of T_{cx} were identified from light scattering signals in conjunction with visual observations as described in the text.

x_{Ani}	T_{cx} (K)
0.1072 ± 0.0004	310.28 ± 0.10
0.1634	322.89 ± 0.02
0.2161	330.22 ± 0.02
0.2742	335.56 ± 0.02
0.3331	338.70 ± 0.02
0.3597	339.70 ± 0.02
0.3933	340.54 ± 0.02
0.4349	341.14 ± 0.02
0.4515	341.29 ± 0.02
0.4872	341.44 ± 0.02
0.4909	341.45 ± 0.02
0.5194	341.49 ± 0.02
0.5310	341.47 ± 0.02
0.5728	341.44 ± 0.02
0.6076	341.30 ± 0.03
0.6499	340.80 ± 0.05
0.7148	338.92 ± 0.05
0.7646	335.67 ± 0.05
0.8022	331.52 ± 0.05
0.8617	319.95 ± 0.10[a]
0.9308	298 ± 5[a]

[a]Measurement based solely on visual observations.

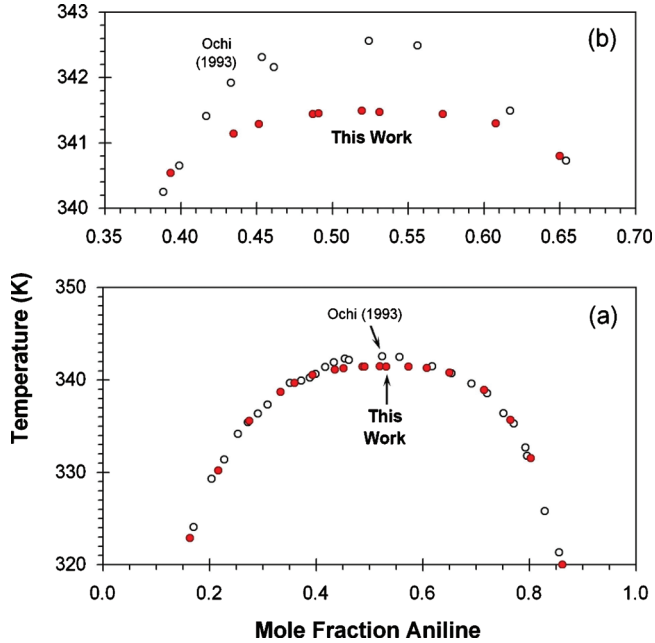

FIG. 8. (a) Comparison of the coexistence curves for the aniline+hexane binary liquid system: (●) this work; (○) Ochi *et al.* (Ref. 47). (b) Expanded view of the critical region.

tification for applying this technique to transmitted signals that decline with temperature prior to T_{cx} due to OZD scattering, and our data show that the linear intersection method will give incorrect results. For example, with near-critical aniline+hexane samples (Figs. 4 and 7), the intersection would occur tenths of a kelvin higher than the true T_{cx} because the phase transition occurs at the end of the rapid decline in transmitted light.

Finally, for light scattering signals collected off axis, assignments of T_{cx} in the literature have been based on the onset of an increase in collected intensity. Our data support this assignment for samples with $|T'_{cx}| > \approx 0.4$ K/mol %, but again show that it is incorrect for near-critical samples because of OZD scattering (Figs. 4–7). As an illustration, we compare our aniline+hexane coexistence curve data (Table II, Fig. 8) to data reported by Ochi *et al.*[47] These authors collected light scattering at 45° from a sample container with an optical path length of 3.5 cm. We find excellent agreement between our T_{cx} data and their data for samples with either $x_{Ani} < 0.4$ or $x_{Ani} > 0.6$. However, the T_{cx} values reported by Ochi *et al.* in the region of the coexistence curve where $|T'_{cx}| < 0.2$ K/mol % are consistently higher than ours—by more than a degree at the critical point. Inspection of our 24° and 90° data in Figs. 4 and 7 shows that Ochi *et al.*'s values are consistent with assigning T_{cx} to the onset of critical opalescence in the one-phase region rather than to the actual phase transition.

VI. CONCLUSIONS

The construction of instrumentation to automate coexistence curve determination is motivated by the prospect of removing direct human observation from a time-consuming procedure. Success of such instrumentation rests on the assumption that interpretation of the recorded light scattering signals is straightforward and can be carried out with high precision and accuracy. If scattered light arose only from the nucleation and growth of droplets, then this assumption would likely be true. Critical opalescence complicates the issue, as the variation of the light scattering signal with temperature from stirred samples is different for different regions of the coexistence curve.

Our data confirm that the identification of T_{cx} for samples with $|T'_{cx}| > \approx 0.4$ K/mol % is best based on the onset of either (i) the decline in transmitted light intensity or (ii) the increase in small-angle light scattering. For these samples, the determination of T_{cx} is limited by equilibration time and by the precision and accuracy of the temperature probe, rather than by interpretation of the light scattering signal. However, direct visual observation remains a more accurate method for identifying T_{cx} for samples where $|T'_{cx}| < 0.2$ K/mol %. Contributions from OZD scattering preclude the presence of a unique signature for the phase transition in light scattering data from these stirred samples. Establishment of a simple rule of interpretation is not possible because the nature of the scattering signal depends on (i) the specific system, (ii) the specific composition of a sample from that system, and (iii) the instrumental path length. Based upon our data, we estimate that examination of *just the light scattering data* from near-critical samples provides an accuracy in T_{cx} of at best ± 0.05 K when a combination of transmitted, 2°, and 90° scattering data are all available for comparison. Efforts are currently underway in our laboratory to improve upon the accuracy with which T_{cx} values of near-critical samples can be assigned from the light scattering signals collected with semiautomated instrumentation.

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APPENDIX: THE LEVER RULE AND RATE OF PHASE FORMATION

When a two-component system is in the two-phase region, the ratio of the total moles n_α in phase α to the total moles n_β in phase β is given by the lever rule,[91]

$$\frac{n_\alpha}{n_\beta} = \frac{x_\beta - x_{\text{sys}}}{x_{\text{sys}} - x_\alpha}. \tag{A1}$$

In this equation, x_α is the composition of phase α, x_β is the composition of phase β, and x_{sys} is the overall composition of the system. Now consider a single-phase system at equilibrium on the coexistence curve with a composition removed from the critical composition. A temperature jump across the coexistence curve causes the spontaneous formation of a new phase α from the mother phase β. A small temperature jump $\mathrm{d}T$ will create $\mathrm{d}n_\alpha$ moles of the new phase. The total moles n_β of the mother phase will not change significantly after such a small jump, and neither will the difference between the system composition and the composition of the new phase, $x_{\text{sys}} - x_\alpha$. With both denominators in Eq. (A1) effectively held constant, the moles $\mathrm{d}n_\alpha$ of the new phase will be proportional to the magnitude of $\mathrm{d}x_\beta$, the displacement of the new mother phase composition from the system composition,

$$\mathrm{d}n_\alpha \propto |x_\beta - x_{\text{sys}}| = |\mathrm{d}x_\beta|. \tag{A2}$$

Equation (A2) leads to the following relationship regarding the rate of change in moles of the new phase α with respect to temperature:

$$\frac{\mathrm{d}n_\alpha}{|\mathrm{d}T|} \propto \left|\frac{\mathrm{d}x_\beta}{\mathrm{d}T}\right| = 1/\left|\frac{\mathrm{d}T}{\mathrm{d}x_\beta}\right|. \tag{A3}$$

Equation (A3) shows that the rate at which moles of α form as the temperature changes is inversely proportional to the slope of the coexistence curve on the mother-phase side of the phase diagram, nearest the isopleth.

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